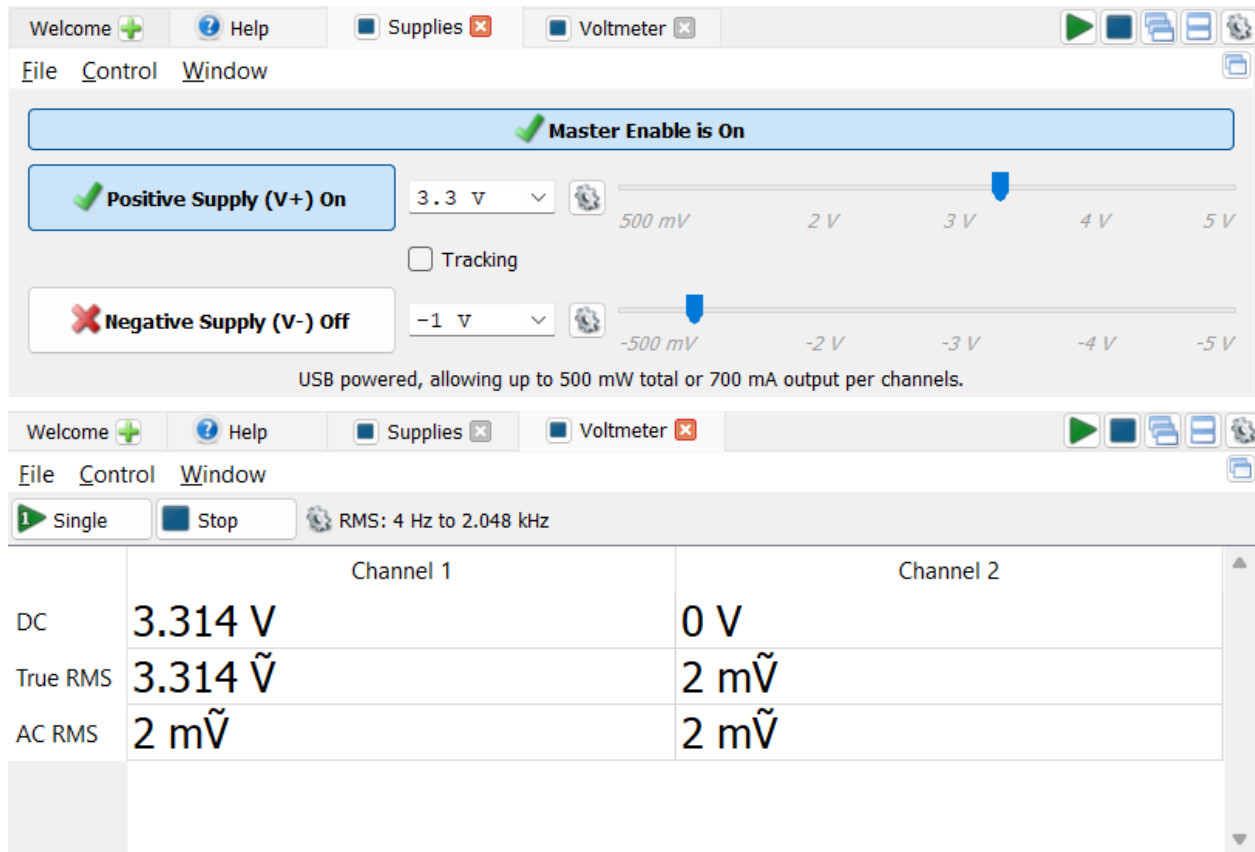
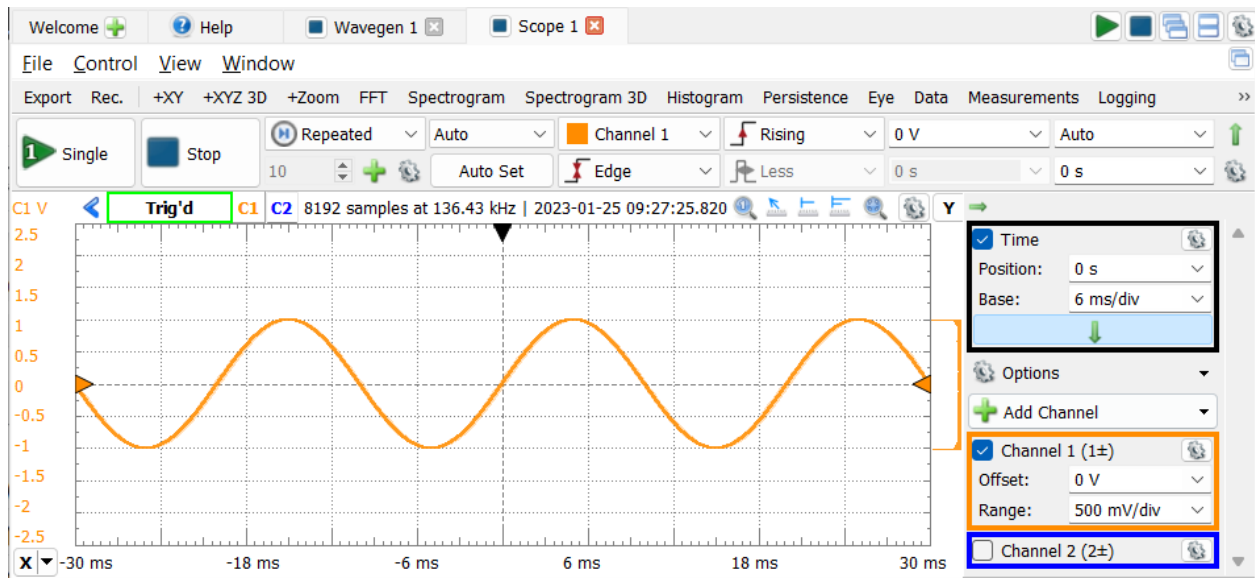
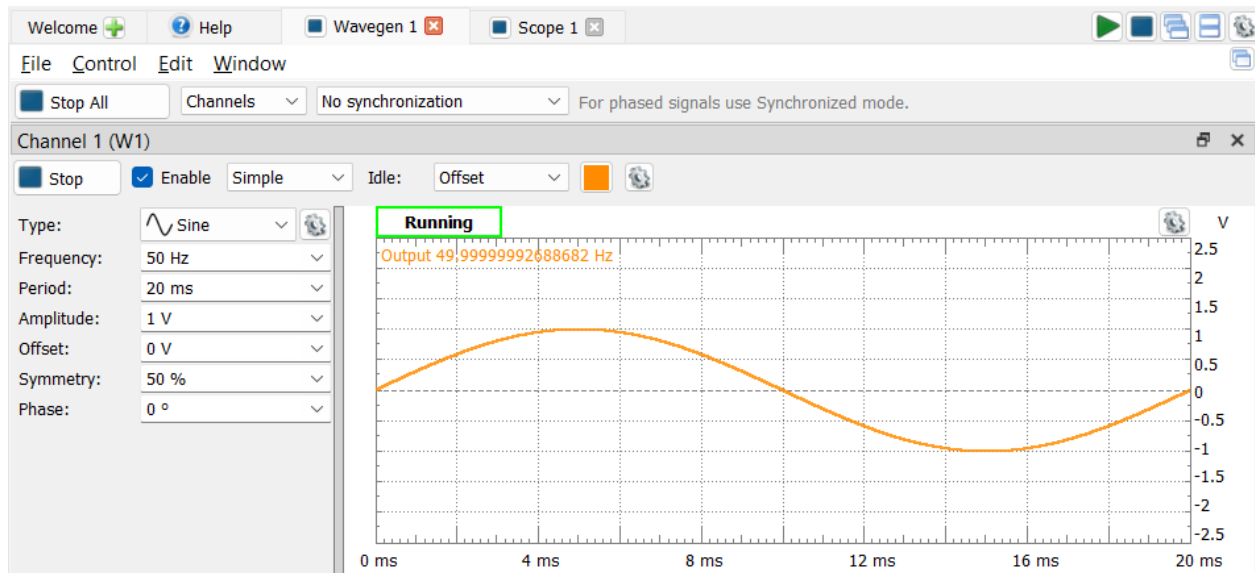


Hayden Fuller
Generate and measure directly:
DC:



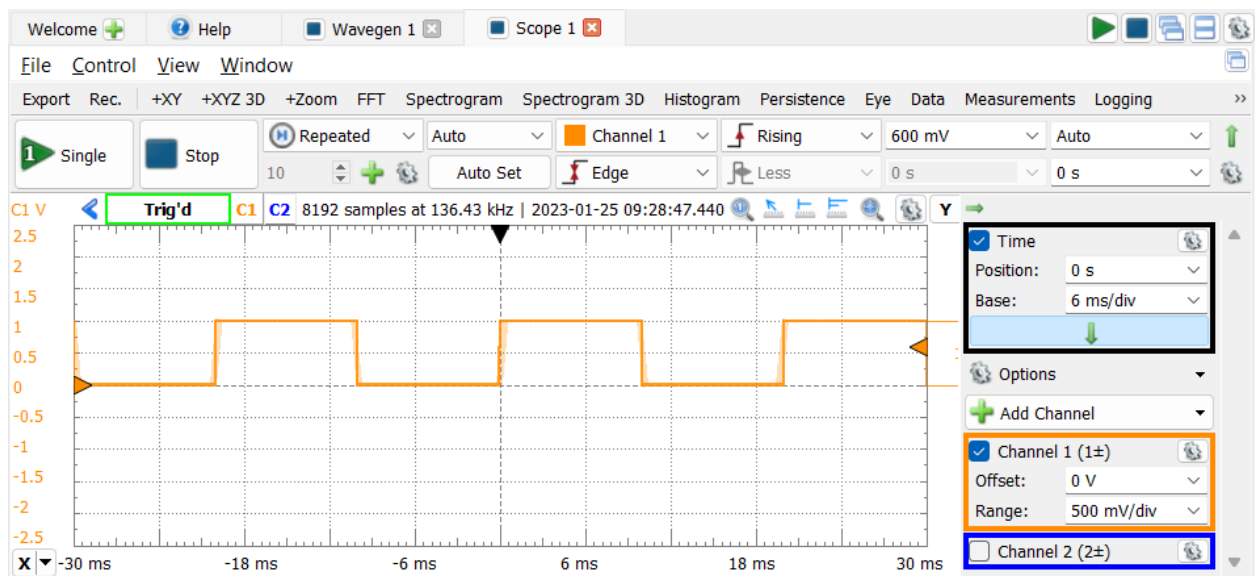
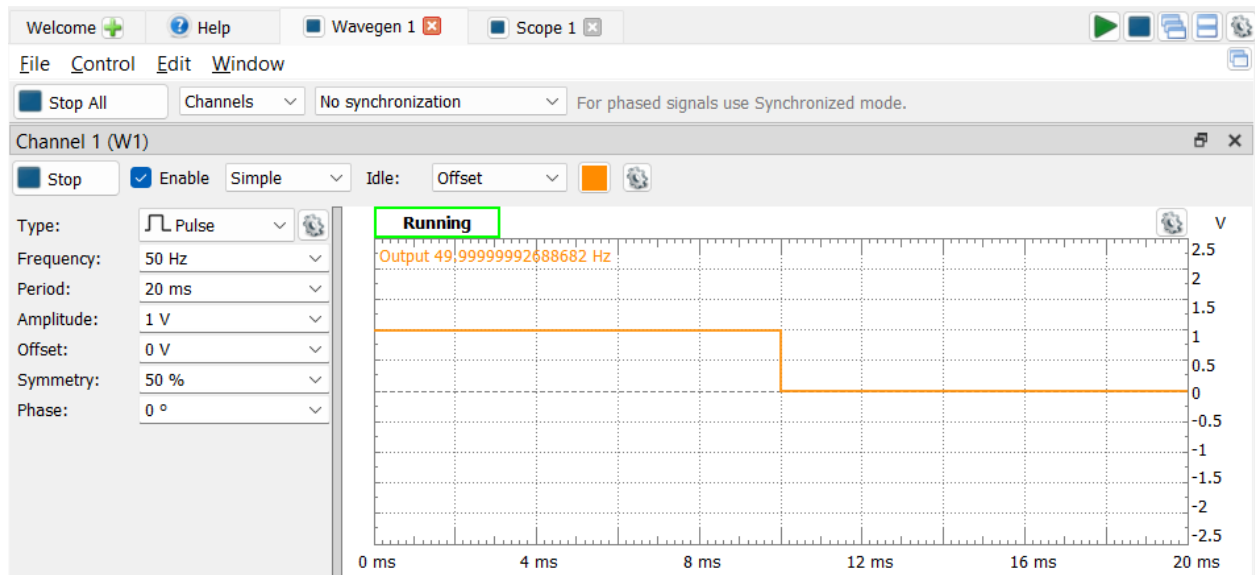
Sinusoid:

W1 — **1+**

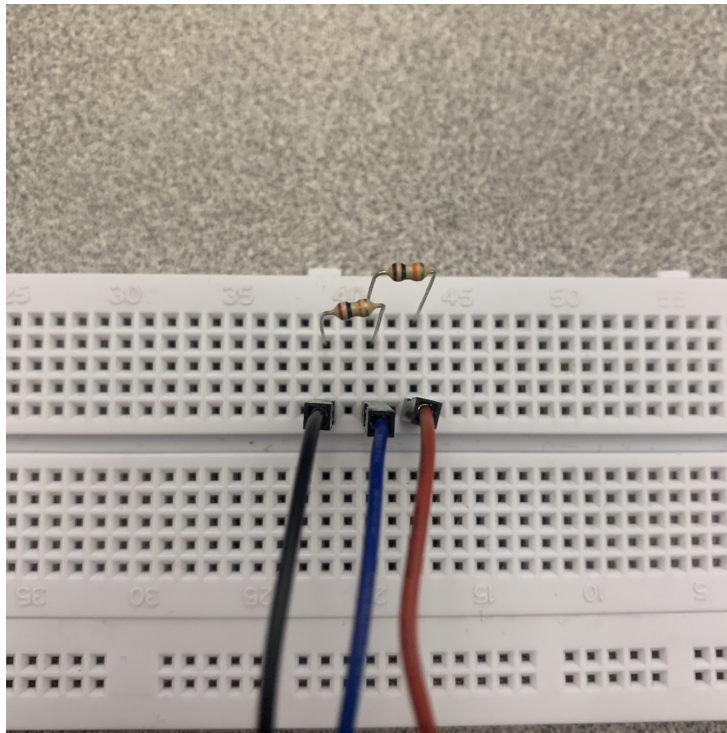
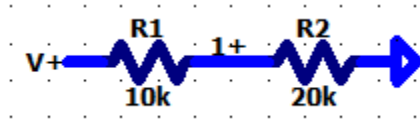


Pulsed signal:

W1 ————— **1+**



Measure DC across one resistor: Simple 10k and 20k in series (20k on gnd side which will be measured across for the remaining objectives)

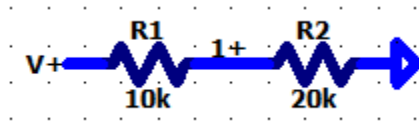


5v:

Red: V+, 5V

Black: Gnd

Blue: 1+, Channel 1



Welcome Help Supplies Voltmeter

File Control Window

Master Enable is On

Positive Supply (V+) On 5 V 500 mV 2 V 3 V 4 V 5 V

☐ Tracking

Negative Supply (V-) Off -5 V -500 mV -2 V -3 V -4 V -5 V

USB powered, allowing up to 500 mW total or 700 mA output per channels.

☐ System Monitor

Welcome Help Supplies Voltmeter

File Control Window

Single Stop RMS: 4 Hz to 2.048 kHz

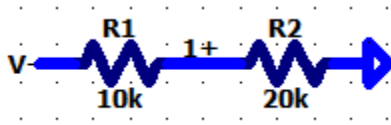
	Channel 1	Channel 2
DC	3.33 V	0 V
True RMS	3.33 V	2 mV
AC RMS	2 mV	2 mV

-5V:

Red: V-, -5V

Black: Gnd

Blue: 1+, Channel 1



Welcome + Help Supplies Voltmeter

File Control Window

Master Enable is On

Positive Supply (V+) Off 5 V 500 mV 2 V 3 V 4 V 5 V

Tracking

Negative Supply (V-) On -5 V -500 mV -2 V -3 V -4 V -5 V

USB powered, allowing up to 500 mW total or 700 mA output per channels.

System Monitor

Welcome + Help Supplies Voltmeter

File Control Window

Single Stop RMS: 4 Hz to 2.048 kHz

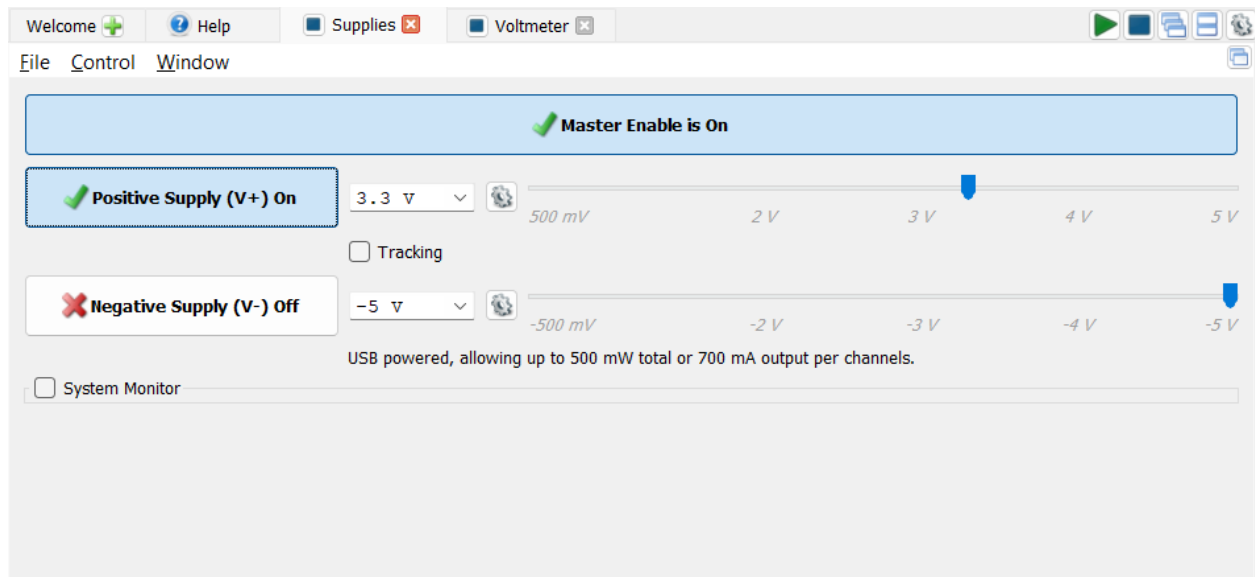
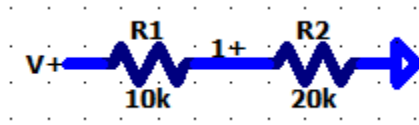
	Channel 1	Channel 2
DC	-3.306 V	2 mV
True RMS	3.306 V	2 mV
AC RMS	2 mV	0 V

3.3V:

Red: V+, 3.3V

Black: Gnd

Blue: 1+, Channel 1



LabVIEW 'Voltmeter' control panel showing measurement results for Channel 1 and Channel 2. The 'Single' button is selected. The 'RMS' checkbox is checked, and the frequency range is set to 4 Hz to 2.048 kHz.

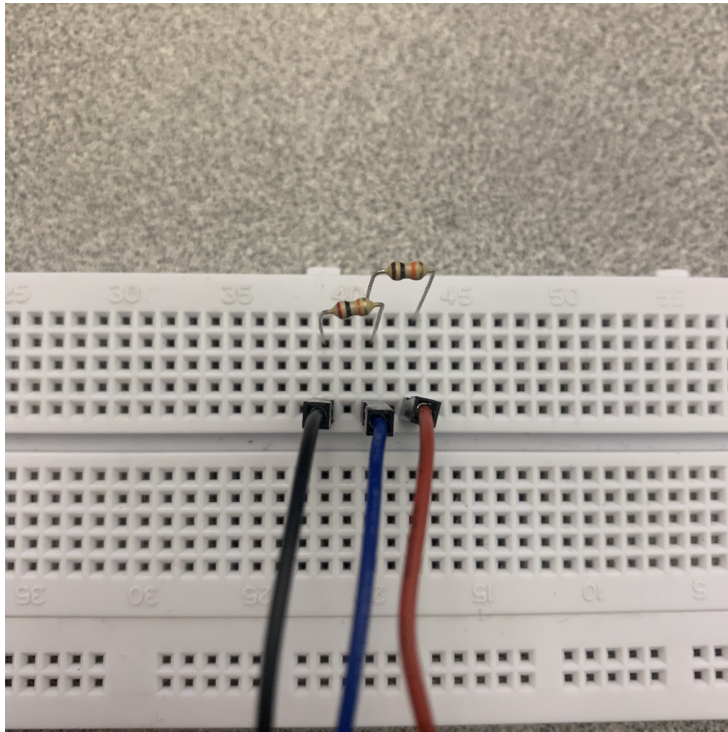
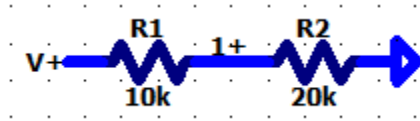
	Channel 1	Channel 2
DC	2.2 V	2 mV
True RMS	2.2 Ṽ	2 mṼ
AC RMS	2 mṼ	0 Ṽ

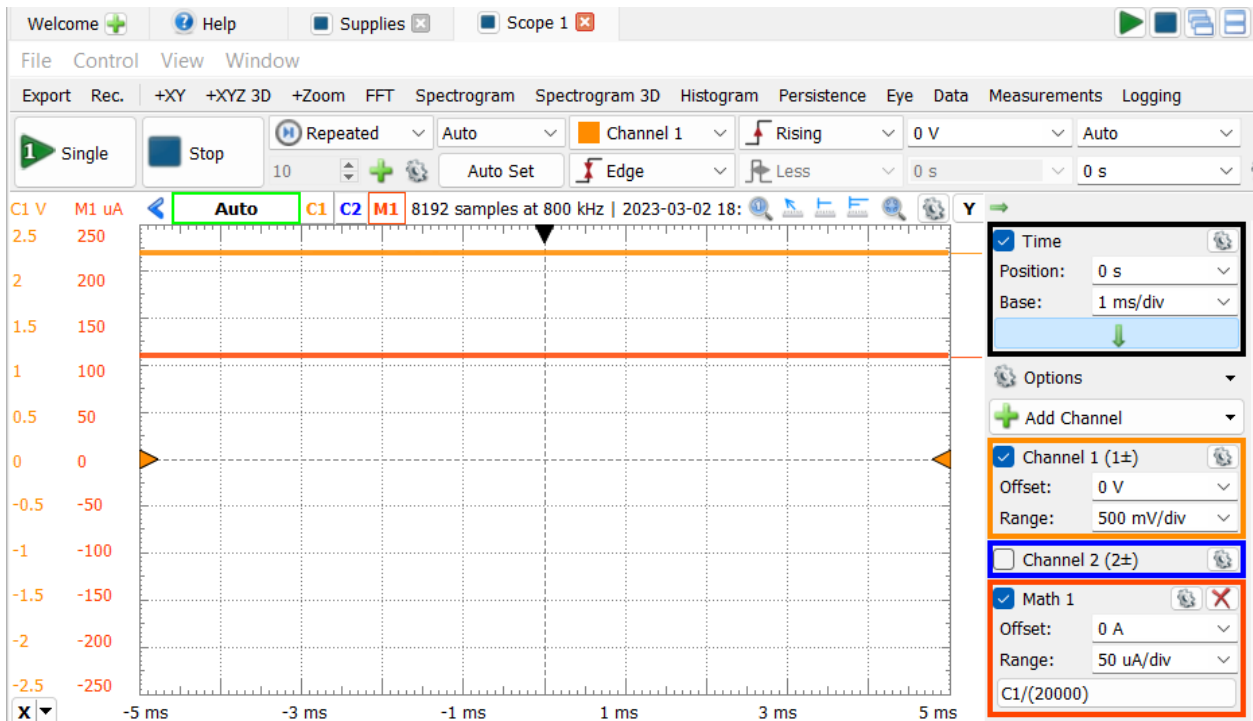
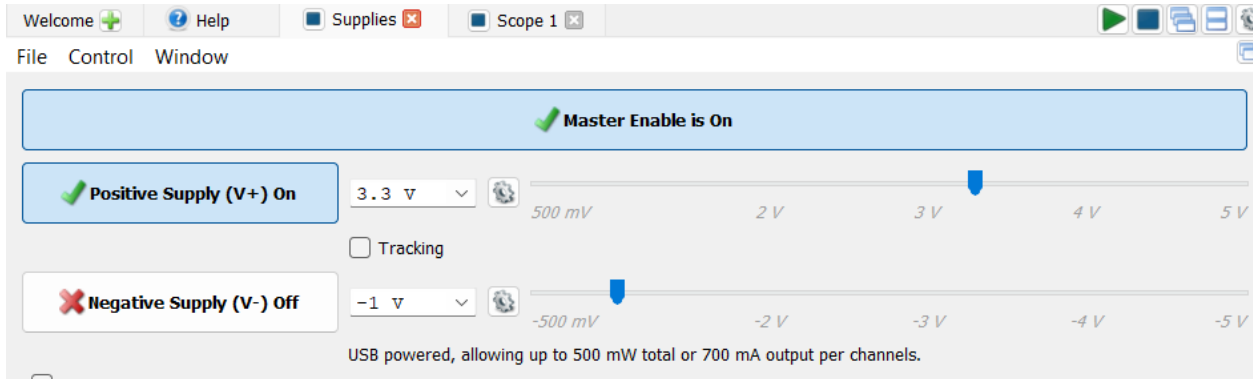
Measure DC current through a resistor: can't be done directly with AD2, used $I=V/R$ with a math channel to calculate it in real time

Red: V+, 3.3V

Black: Gnd

Blue: 1+, Channel 1





I had points taken off for the background not being white. Assuming my vision is correct, that is white.

I also had points taken off for not increasing the line width, but it's also already at the maximum setting.

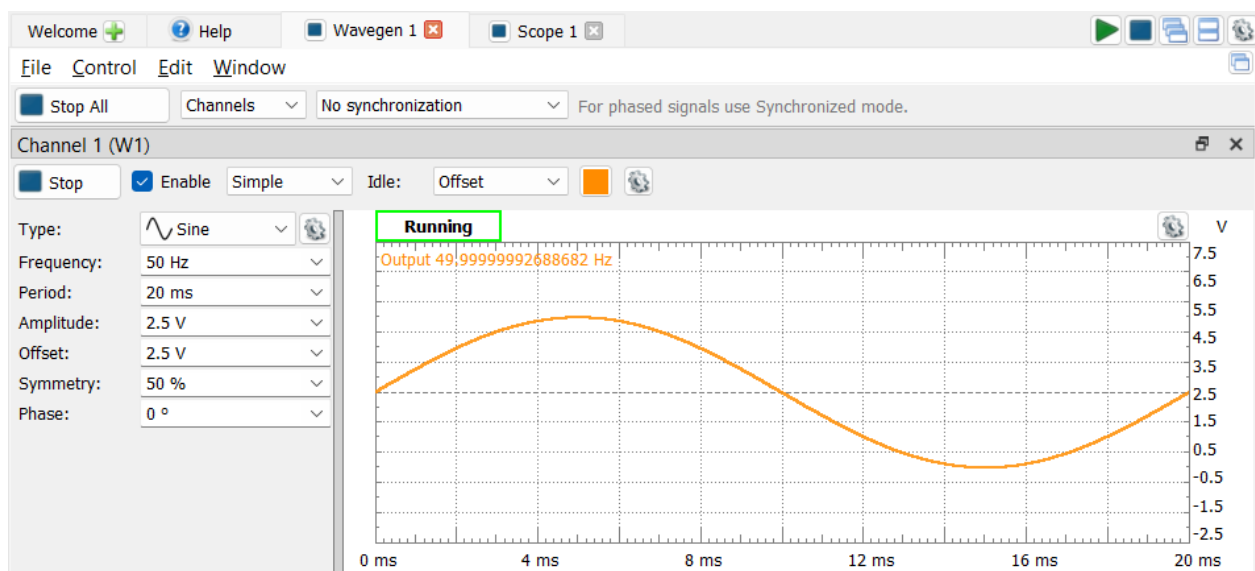
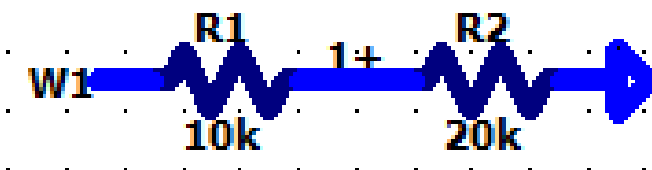
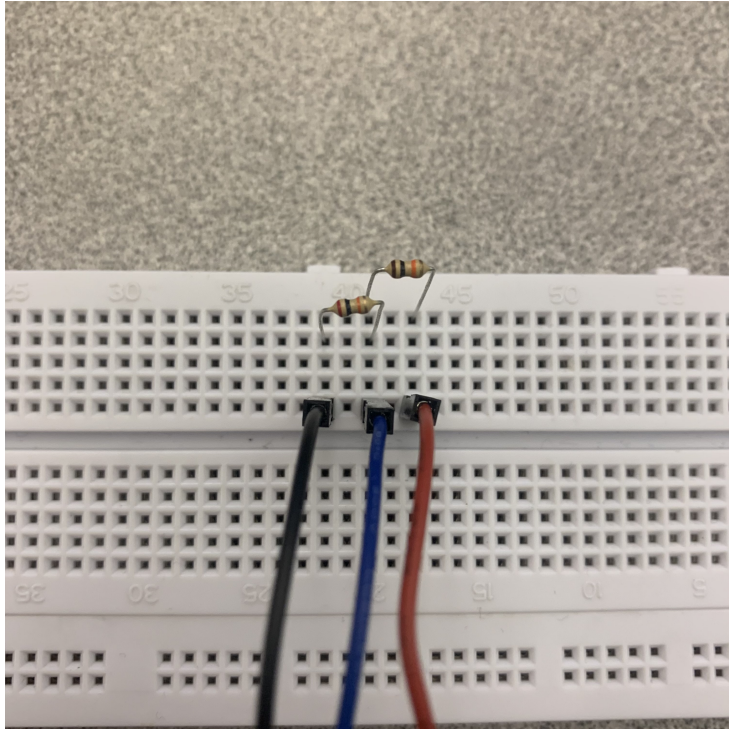
Measure voltage across one resistor with a sinusoid source:

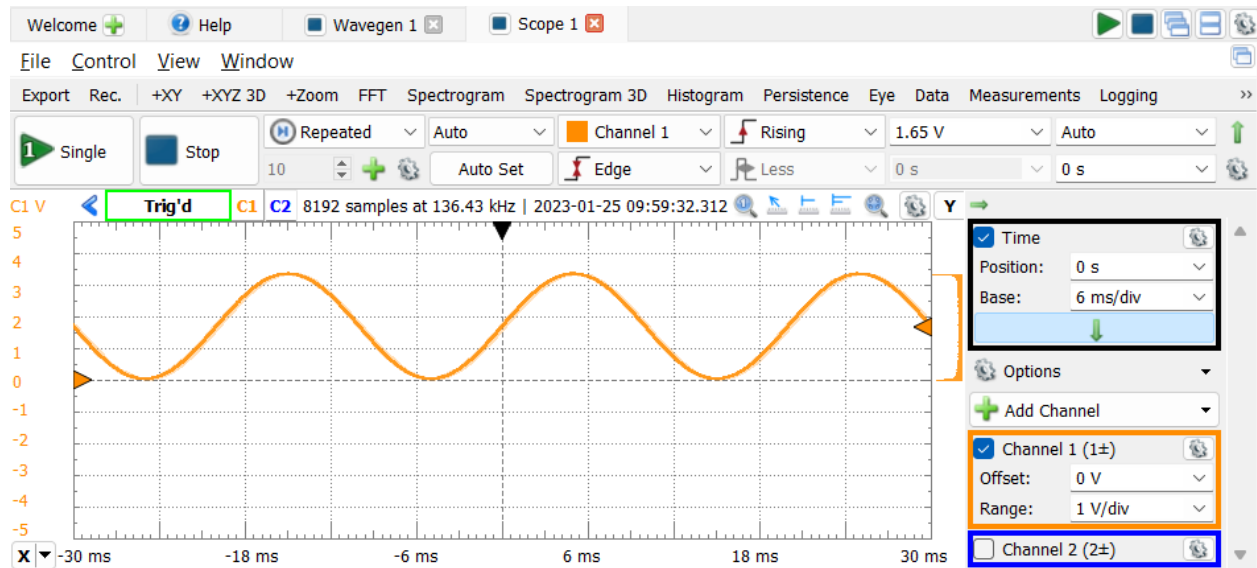
5 to 0:

Red: W1

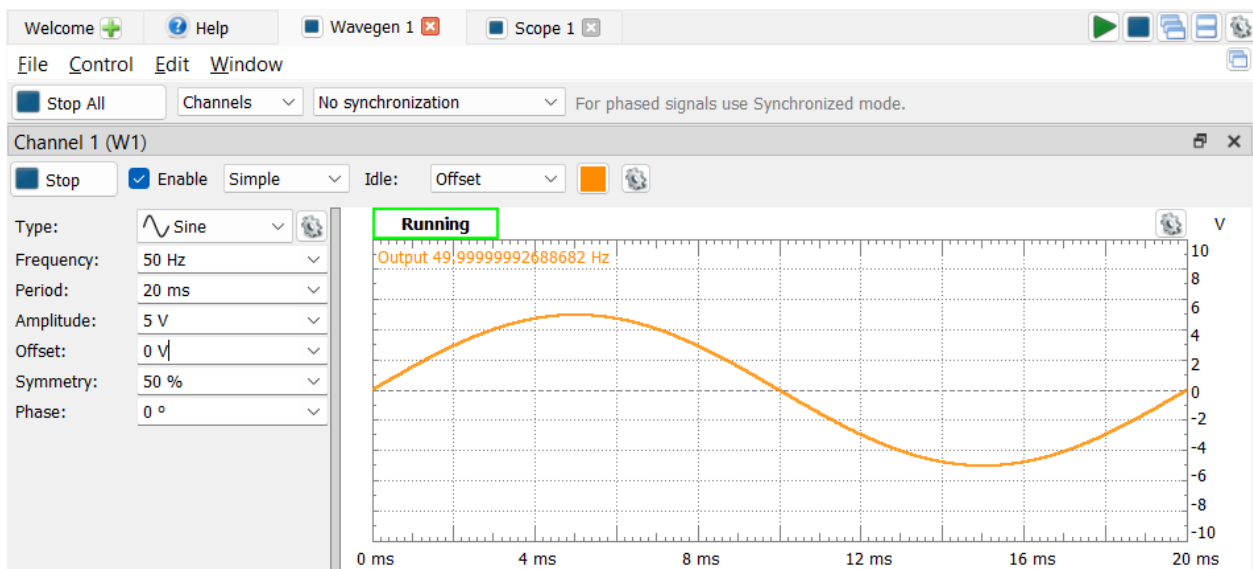
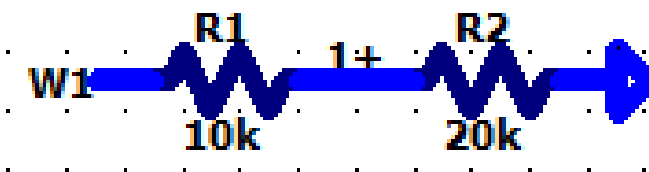
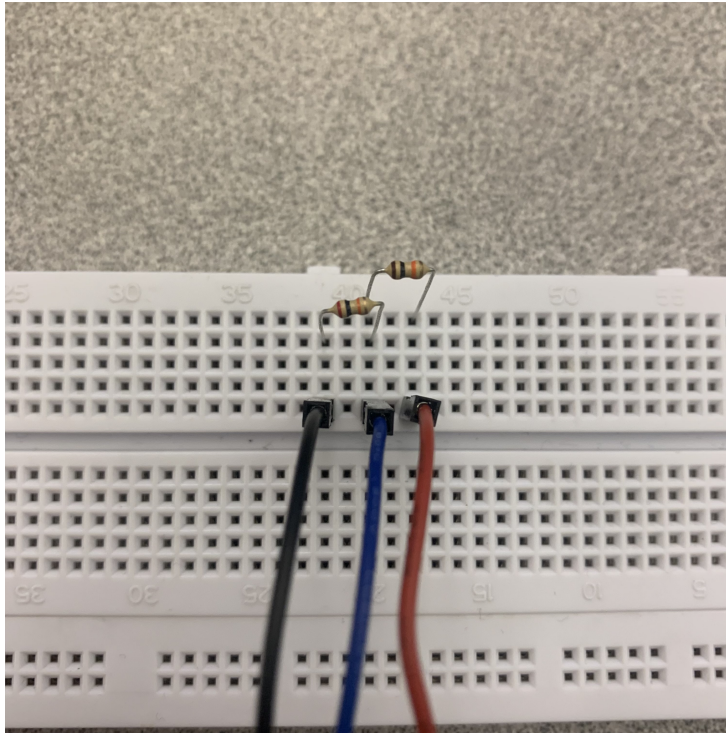
Black: Gnd

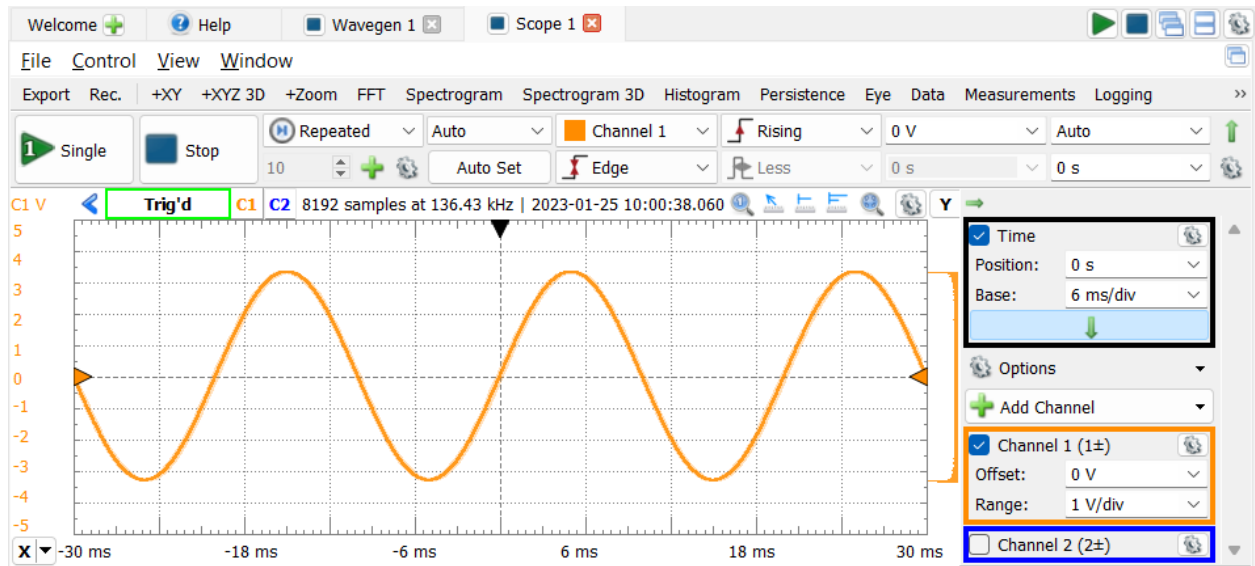
Blue: 1+, Channel 1





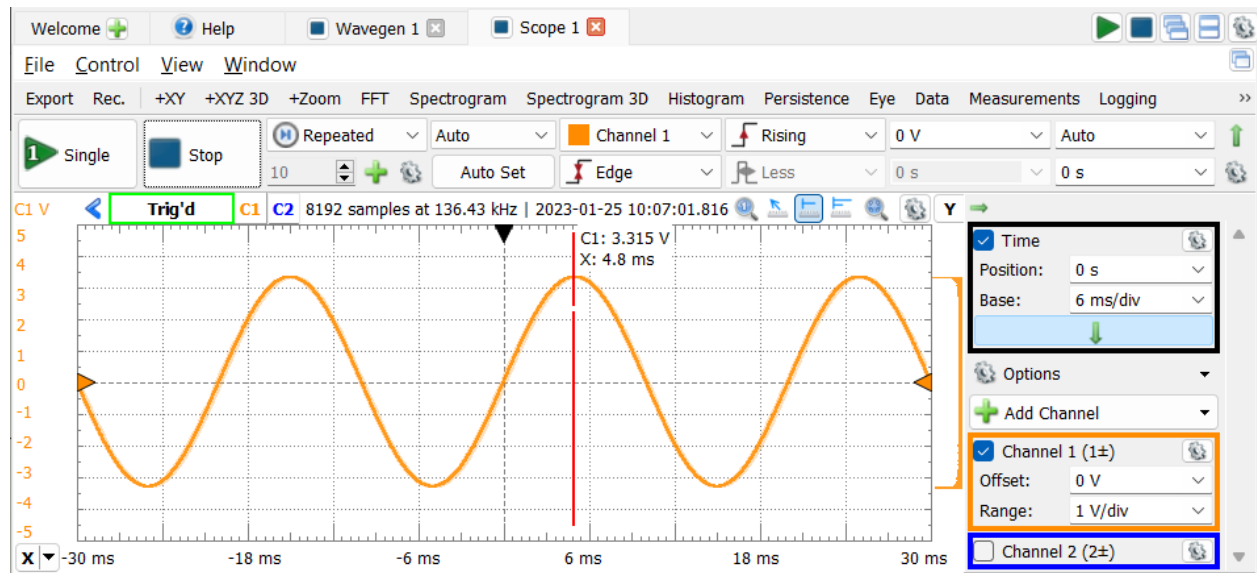
Blue: 1+, Channel 1



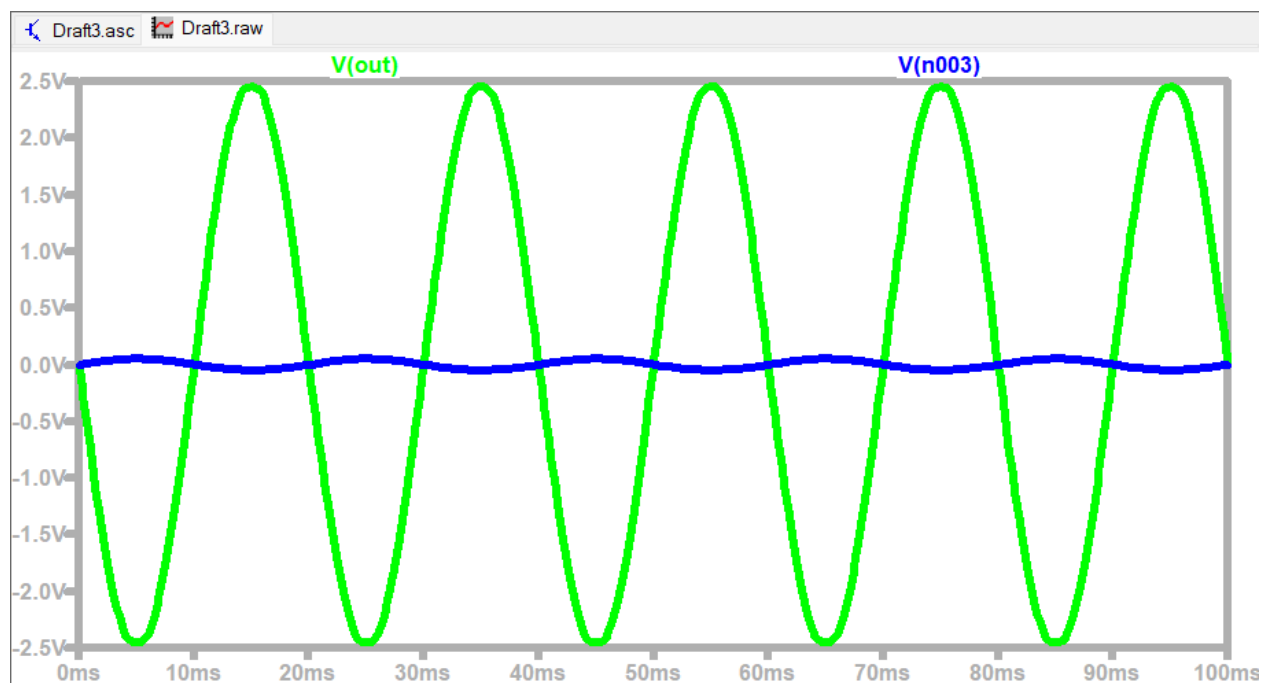
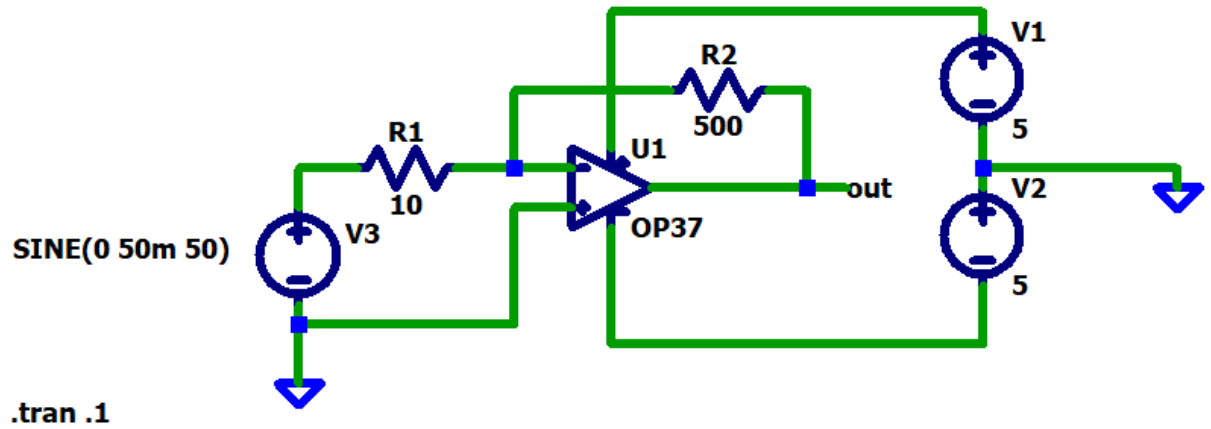


The analog discovery 2 can do both.

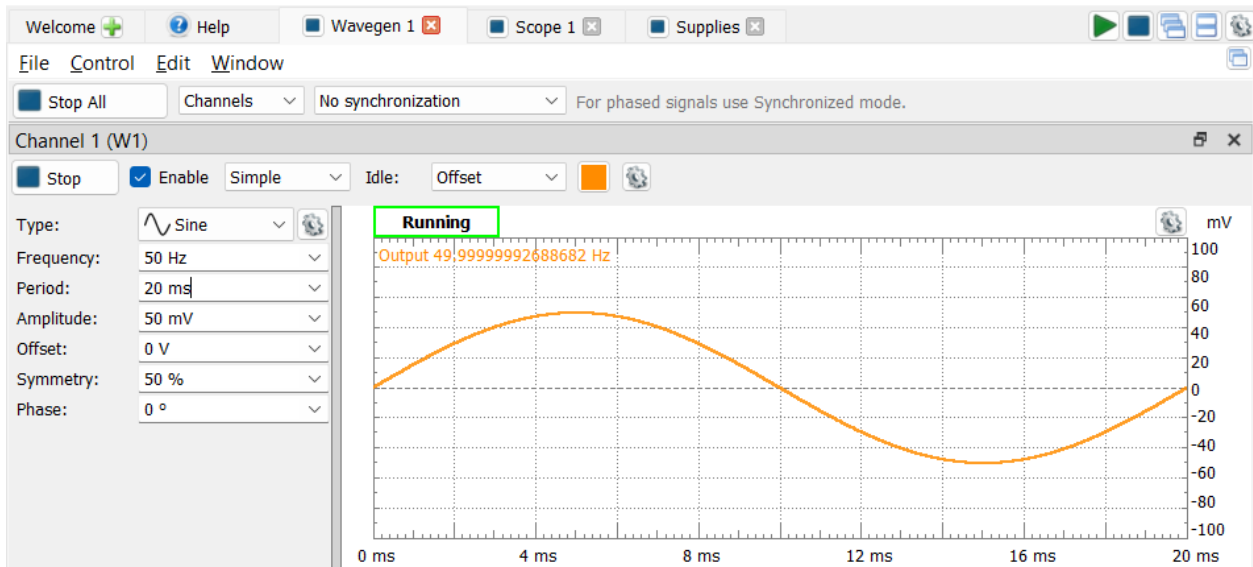
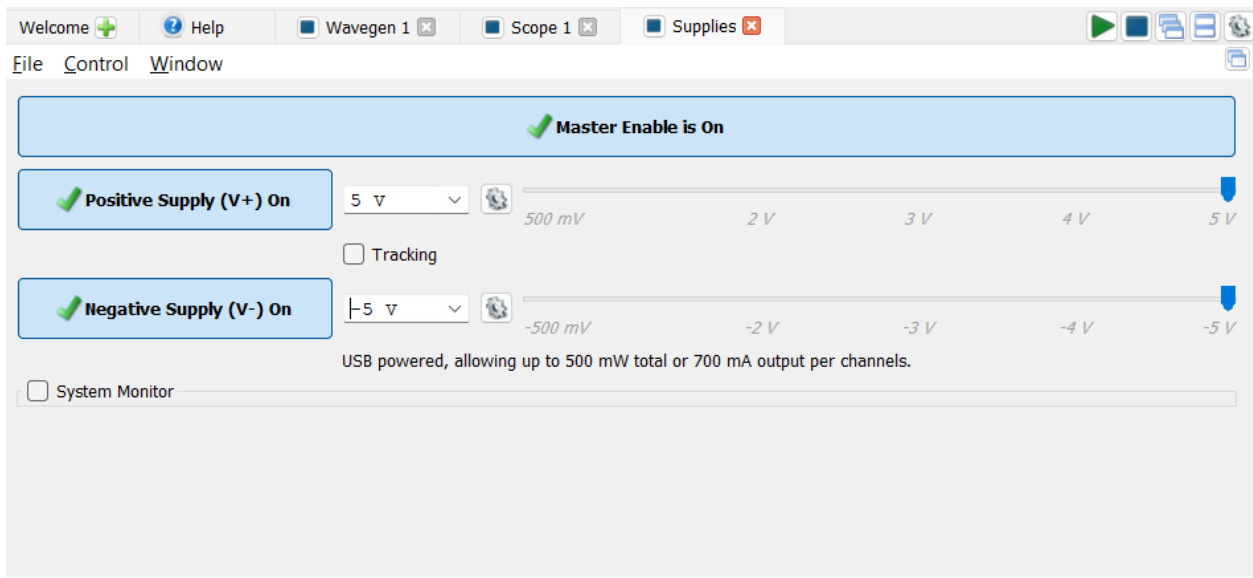
Use cursor function:



Op amp: Inverting amplifier with a gain of 50
LT spice circuit and simulation



Waveforms setup



output

